

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering

CSE 3102, Lab No. 10 , Lab Work

Consider the following instructions. Execute each one and write a report on the answers to the instructions. First complete the following tasks to create the database and the tables.

Task 1: Database and Table Creation

1. Create a database said "HospitalDB"

2. Create a table titled "Doctors" with these fields and constraints:

Field	Type	Constraints
doctor_id	INT	PRIMARY KEY, AUTO_INCREMENT
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
specialization	VARCHAR(100)	NOT NULL
contact_no	VARCHAR(15)	UNIQUE
Joining_date	DATE	NOT NULL

3. Create a table titled "Patients" with these fields and constraints:

Field	Type	Constraints
patient_id	INT	PRIMARY KEY, AUTO_INCREMENT
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
dob	VARCHAR(100)	NOT NULL
gender	ENUM('M', 'F', 'O')	NOT NULL
blood_group	VARCHAR(5)	NULL

4. Create a table titled "Appointments" with these fields and constraints:

Field	Type	Constraints
app_id	INT	PRIMARY KEY, AUTO_INCREMENT
patient_id	INT	FOREIGN KEY REFERENCES Patients(patient_id)
doctor_id	INT	FOREIGN KEY REFERENCES Doctors(doctor_id)
app_date	DATETIME	NOT NULL
status	ENUM('Pending', 'Completed', 'Cancelled')	DEFAULT 'Pending'

Task 2: Data Insertion

5. Insert these records into the Doctors table:

first_name	last_name	specialization	contact_no	joining_date
Dr. John	Smith	Cardiology	'01711223344'	'2015-06-10'
Dr. Sarah	Lee	Pediatrics	'01755667788'	'2018-03-15'
Dr. Robert	Brown	Neurology	'01799887766'	'2020-11-20'

6. Insert these records into the Patients table:

first_name	last_name	dob	gender	blood_group
Michael	Johnson	'1985-08-12'	'M'	'A+'
Emily	Davis	'1992-03-25'	'F'	'B-'
David	Wilson	'1978-11-30'	'M'	'O+'

7. Insert these records into the Appointments table:

patient_id	doctor_id	app_date	status
1	1	'2023-10-15 10:00:00'	'Completed'
2	2	'2023-10-16 14:30:00'	'Pending'
3	1	'2023-10-17 09:15:00'	'Pending'

Task 3: SQL Queries

Write SQL queries for the following:

- Display all doctors with their specialization and contact number
- List all patients with their age (calculated from DOB)
- Find all appointments scheduled for October 2023
- Count the number of patients by blood group
- List all pending appointments with patient and doctor names
- Find doctors who joined after 2018
- Calculate the total appointments handled by each doctor
- Find patients with blood group 'A+'
- List all completed appointments with patient details
- Find the doctor with the most appointments

Query Hints:

Query	SQL Concepts Required
a	Basic SELECT, CONCAT()
b	DATEDIFF(), YEAR()
c	BETWEEN, DATE functions
d	GROUP BY, COUNT()
e	INNER JOIN, WHERE
f	DATE comparison
g	JOIN with GROUP BY
h	Simple WHERE filter
i	Multiple table JOIN
j	Subquery with MAX()

Answer:

```
CREATE DATABASE HospitalDB;  
USE HospitalDB;
```

```
CREATE TABLE Doctors (  
    doctor_id INT PRIMARY KEY AUTO_INCREMENT,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    specialization VARCHAR(100) NOT NULL,  
    contact_no VARCHAR(15) UNIQUE,  
    joining_date DATE NOT NULL  
);
```

```
CREATE TABLE Patients (  
    patient_id INT PRIMARY KEY AUTO_INCREMENT,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    dob DATE NOT NULL,  
    gender ENUM('M', 'F', 'O') NOT NULL,  
    blood_group VARCHAR(5)  
);
```

```
CREATE TABLE Appointments (  
    app_id INT PRIMARY KEY AUTO_INCREMENT,  
    patient_id INT,  
    doctor_id INT,  
    app_date DATETIME NOT NULL,  
    status ENUM('Pending', 'Completed', 'Cancelled') DEFAULT 'Pending',  
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)  
);
```

```
INSERT INTO Doctors (first_name, last_name, specialization, contact_no, joining_date)  
VALUES  
( 'John', 'Smith', 'Cardiology', '01711223344', '2015-06-10'),  
( 'Sarah', 'Lee', 'Pediatrics', '01755667788', '2018-03-15'),  
( 'Robert', 'Brown', 'Neurology', '01799887766', '2020-11-20');
```

- **INSERT INTO** Patients (first_name, last_name, dob, gender, blood_group)
VALUES
('Michael', 'Johnson', '1985-08-12', 'M', 'A+'),
('Emily', 'Davis', '1992-03-25', 'F', 'B-'),
('David', 'Wilson', '1978-11-30', 'M', 'O+');
- **INSERT INTO** Appointments (patient_id, doctor_id, app_date, **status**)
VALUES
(1, 1, '2023-10-15 10:00:00', 'Completed'),
(2, 2, '2023-10-16 14:30:00', 'Pending'),
(3, 1, '2023-10-17 09:15:00', 'Pending');

-- Query-01 Display all doctors with their full names, specialization, and contact numbers

SELECT

doctor_id,

CONCAT('Dr. ', first_name, ' ', last_name) AS full_name,

specialization,

contact_no

FROM Doctors;

Output:

	doctor_id	full_name	specialization	contact_no
▶	1	Dr. John Smith	Cardiology	01711223344
	2	Dr. Sarah Lee	Pediatrics	01755667788
	3	Dr. Robert Brown	Neurology	01799887766

-- Query-02 List all patients with their full names and calculated age

```
SELECT
    patient_id,
    CONCAT(first_name, ' ', last_name) AS full_name,
    TIMESTAMPDIFF(YEAR, dob, CURDATE()) AS age
FROM Patients;
```

Output:

	patient_id	full_name	age
▶	1	Michael Johnson	39
	2	Emily Davis	33
	3	David Wilson	46

-- Query-03 Find all appointments scheduled for October 2023

```
SELECT
    a.app_id,
    CONCAT(p.first_name, ' ', p.last_name) AS patient_name,
    CONCAT(d.first_name, ' ', d.last_name) AS doctor_name,
    a.app_date,
    a.status
FROM Appointments a
JOIN Patients p ON a.patient_id = p.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE a.app_date BETWEEN '2023-10-01' AND '2023-10-31';
```

Output:

	app_id	patient_name	doctor_name	app_date	status
▶	1	Michael Johnson	John Smith	2023-10-15 10:00:00	Completed
	2	Emily Davis	Sarah Lee	2023-10-16 14:30:00	Pending
	3	David Wilson	John Smith	2023-10-17 09:15:00	Pending

-- Query-04 Count the number of patients by blood group

```
SELECT
    blood_group,
    COUNT(*) AS patient_count
FROM Patients
WHERE blood_group IS NOT NULL
GROUP BY blood_group;
```

Output:

	blood_group	patient_count
▶	A+	1
	B-	1
	O+	1

-- Query-05 List all pending appointments with patient and doctor names

```
SELECT
    a.app_id,
    CONCAT(p.first_name, ' ', p.last_name) AS patient_name,
    CONCAT(d.first_name, ' ', d.last_name) AS doctor_name,
    a.app_date
FROM Appointments a
JOIN Patients p ON a.patient_id = p.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE a.status = 'Pending';
```

Output:

	app_id	patient_name	doctor_name	app_date
▶	2	Emily Davis	Sarah Lee	2023-10-16 14:30:00
	3	David Wilson	John Smith	2023-10-17 09:15:00

-- Query-06 Find doctors who joined after 2018

```
SELECT
    doctor_id,
    CONCAT(first_name, ' ', last_name) AS full_name,
    joining_date
FROM Doctors
WHERE YEAR(joining_date) > 2018;
```

Output:

	doctor_id	full_name	joining_date
▶	3	Robert Brown	2020-11-20

-- Query-07 Calculate the total appointments handled by each doctor

```
SELECT
    CONCAT(d.first_name, ' ', d.last_name) AS doctor_name,
    COUNT(a.app_id) AS total_appointments
FROM Doctors d
LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id
GROUP BY d.doctor_id;
```

Output:

	doctor_name	total_appointments
▶	John Smith	2
	Sarah Lee	1
	Robert Brown	0

-- Query-08 Find patients with blood group 'A+'

```
SELECT
    patient_id,
    CONCAT(first_name, ' ', last_name) AS full_name,
    blood_group
FROM Patients
WHERE blood_group = 'A+';
```

Output:

	patient_id	full_name	blood_group
▶	1	Michael Johnson	A+

-- Query-09 List all completed appointments with patient details

```
SELECT
    a.app_id,
    CONCAT(p.first_name, ' ', p.last_name) AS patient_name,
    p.dob,
    p.gender,
    CONCAT(d.first_name, ' ', d.last_name) AS doctor_name,
    a.app_date
FROM Appointments a
JOIN Patients p ON a.patient_id = p.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE a.status = 'Completed';
```

Output:

	app_id	patient_name	dob	gender	doctor_name	app_date
▶	1	Michael Johnson	1985-08-12	M	John Smith	2023-10-15 10:00:00

-- Query-10 Find the doctor with the most appointments

SELECT

CONCAT(d.first_name, ' ', d.last_name) AS doctor_name,

COUNT(a.app_id) AS appointment_count

FROM Doctors d

JOIN Appointments a ON d.doctor_id = a.doctor_id

GROUP BY d.doctor_id

ORDER BY appointment_count DESC

LIMIT 1;

Output:

	doctor_name	appointment_count
▶	John Smith	2